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ICS 2104 Object-Oriented Programming I

This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 4

Friend Functions, Static Functions and Constant Functions

Learning outcomes

Upon completing this topic you should be able to:

- Differentiate a friend function, a static function and a constant function
- Use friend functions, static functions and constant functions

4.1. A Friend Function

A friend function is a function that is not a member of a class (also known as external functions) but it can access data members of the class just like members function of that class.

4.1.1. Making an External Function a Friend Function

An external function is made a friend function by preceding its declaration with friend keyword as shown below:

```
class Account

        private :
                ...
        public:
```

```
        friend void compute_tax(Account);
        . . .
};
```

Note:

- compute_tax() function is not a member function of Account class. The reserved word friend make this function to behavior like a member function of Account class
- it can not be called using an object of Account class. Therefore, it is called just like a normal function.
- Because compute_tax() function is not a member of Account class, it can access Account class data members through the use of Account object. Therefore its argument should be an object of type Account

- compute_tax() can be declared in any section of Account class(private, public and protected sections)

The program below demonstrates the use of friend function

```
/*----------------------------------------------------------------
Example.cpp This program
demonstrates the use of friend functions
---------------------------------------------------------------- *

#include<iostream.h>
class Example
{
        private:
                int x;
                int y;
        public:
```

```
Example( int x1, int y1)
{
        x=x1;
        y=y1;
}
friend void print_value(Example);
};
void print_value(Example e)
{
cout<<The values are <<e.x <<and
<<e.y<<endl;
}
int main(void)
{
```

```
    Example myExample(10, 30);
    print_value(myExample);
    return 0;
}
```

4.1.2. Making a Member Function a Friend Function

There are member functions of a class that are friend to other classes.

That means these member functions are able to access data members of that classes as if they are member functions of those classes.

Such member functions are made friend to class as shown below

```
The values are 10 and 30
Press any key to continue
```

Table 4.1: Output from the above program

```
class class-name1
{
        private:
                    . . .
        public:
                    . . .
        friend return-type class-name2::
function-name();
};
```

Example

```
class B
{
        private:
```

```
        . . .
    public:
        . . .
    friend int A :: y(B b);
};
```

where

- friend is a reserved word
- int is the return-type of the function y()
- A is the class that y() function is a member of
- :: scope resolution operators
- y() is the function that is being made a friend of B class.
- y() function behavior as if is a class B member function.

4.1.3. Making all Member Functions to be Friends

All the member functions of a class can be made to be friend to another class.

For example:

```
class B {
private:
. . .
public:
. . .
friend class A ;
};
```

where all A class member functions are now friends to class B

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4.2. const member function

A const member function is a member function that does not modify data members in a class.

The reserved word const is appended at the end of the function declaration and function definition as shown below

```
//function declaration
void func() const;
//function definitions
void func() const
{
function body
}
```

4.3. static member function

A static member function is a member function that has been declared with the static keyword.

A static member function can access only a static variable and it is called using the name of class separated with scope resolution operators.

The syntax of declaring and calling a static member function is

a) declaring a class with static member function

```
class class-name
{
        private:
                . . .
```

```
        public :
                . . .
        static return-type function-name()
        {
                function body
        }
};
```

b) syntax of calling a static member

```
return-type function-name()
{
        class-name:: static function-name();
}
```

Revision Questions

EXERCISE 1. ✍ Explain why friend functions are passed object of the class that they have been to be a friend of

Example ✎. Differentiate a member function and a friend function

Solution: A member function can access the data members of a class without using an object while a friend function can only use an object to be able to access the data member. □

EXERCISE 2. ✍ Explain any FOUR characteristics of friend functions

EXERCISE 3. ✍ Show how a member function can be made a friend function

EXERCISE 4. ✍ Differentiate static member functions and constant menber functions

References and Additional Reading Materials

- David Flanagan, 2005, Java in a Nutshell (5th Edition), O'Reilly Press,
- Matt Weisfeld, 2009, The Object-Oriented Thought Process, 3rd Edition, , Addison-Wesley,
- Malik, D. S. 2002, C++ Programming: From Problem Analysis to Program Design, Course Technology, Canada
- Pappas, H. Chris and Murray, William, H. 1996, C/C++ Programmer's Guide, BPB Publications, New Delhi
- Flanagan, D. (2005). Java in a nutshell : a desktop quick reference. O'Reilly (5th ed.).
- Flanagan, D. (2004). Java examples in a nutshell : a tutorial companion to Java in a nutshell. O'Reilly (3rd ed.)

Solutions to Exercises

Exercise 1.

The object enables friend function to access the data members of the class.

Exercise 1